The Standard Genetic Code Is Not Locally Optimal Under Simple Chemical Distance

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Abstract

We evaluate the local optimality of the standard genetic code under a simple chemical-distance metric that measures the physicochemical cost of amino acid substitutions caused by single-nucleotide mutations. Using an exhaustive pairwise-swap search over all $\binom{20}{2} = 190$ amino-acid transpositions within the degeneracy-preserving assignment space, we find that the standard code is **not** a local minimum: at least 42 pairwise swaps strictly reduce total weighted mutation cost. The largest improving swap (Arg $\leftrightarrow$ Gln) reduces cost by 10.26%. At the same time, a degeneracy-preserving random baseline of 10,000 shuffled codes shows that the standard code still outperforms 99.47% of sampled alternatives under this metric. The code is therefore globally strong relative to random codes but not locally optimal in its immediate swap neighborhood. These results are formalized in Lean 4 with a machine-checked proof that the standard code is not a local minimum. We conclude that any serious optimality claim for the genetic code must account for structure beyond simple physicochemical substitution cost, such as position-dependent mutation effects, biosynthetic pathway constraints, and translation-machinery compatibility.

1 Introduction

The standard genetic code maps 61 sense codons to 20 amino acids via a famously non-random pattern [8, 1]. Since at least Haig and Hurst [4], a substantial literature has asked whether this mapping is optimal with respect to minimizing the deleterious effects of point mutations. The prevailing view, strengthened by the influential work of Freeland and Hurst [2], is that the standard code is near-optimal: it is better than the vast majority of randomly generated alternatives at minimizing a weighted chemical-distance objective.

This near-optimality claim, however, has always been stated relative to random codes. The question of *local* optimality—whether any small perturbation of the code can strictly improve the objective—has received less attention. In this paper, we address this question directly.

We define a simple chemical-distance metric based on four amino acid physicochemical properties (molecular weight, hydrophobicity, charge, and side-chain volume) and evaluate all 190 pairwise amino-acid-label swaps within the degeneracy-preserving assignment space. We find that **42 out of 190 swaps strictly reduce the total weighted mutation cost**. The standard code is therefore not a local minimum under this metric.

This is not a refutation of the genetic code's optimality. Rather, it is a precise characterization of what a simple metric can and cannot explain, and a motivation for identifying the richer objective that the code actually minimizes.

The central point is slightly subtler, and stronger, than a bare negative result. Under this chemistry-only objective the standard code performs extremely well against random degeneracy-preserving shuffles, yet still admits many improving local perturbations. The natural conclusion is not that the code lacks structure, but that the wrong objective is being optimized.

All results are formalized in Lean 4 and verified

by the Lean type checker. The central formal statement is that there exist amino acids $a \neq b$ such that swapping a and b strictly reduces the total code cost under chemistry-only weights. The exact theorem and module are given in the Formal Verification section below. The full formalization is available in the `IndisputableMonolith.Genetics` namespace (49 modules, 8,000+ lines, zero axioms, zero unproved assertions).

2 Definitions

2.1 Chemical Distance

For amino acids a_1, a_2 with physicochemical properties (w_i, h_i, c_i, v_i) denoting molecular weight (Da), Kyte–Doolittle hydrophobicity ($\times 10$, integer), charge at pH 7 ($\{-1, 0, +1\}$), and side-chain volume (Å^3), the chemical distance is

$$d(a_1, a_2) = \Delta w^2 + \Delta h^2 + 100\,\Delta c^2 + \Delta v^2 \quad (1)$$

where $\Delta w = w_1 - w_2$, etc. The charge term carries a factor of 100 to penalize charge-changing substitutions, which are among the most biologically disruptive [3]. This metric is symmetric, non-negative, and zero if and only if $a_1 = a_2$.

2.2 Mutation Model

Single-nucleotide mutations are weighted by a transition–transversion asymmetry: transitions (purine↔purine or pyrimidine↔pyrimidine) receive weight 2, transversions receive weight 1. This reflects the approximately 2:1 transition bias observed in most genomes [6]. Same-nucleotide "mutations" carry weight 0.

For a codon c with single-nucleotide mutants $\{m_1, \ldots, m_9\}$, the **vulnerability** of c under assignment A is

$$V_A(c) = \sum_{i=1}^{9} \mu(c, m_i)\, K_A(c, m_i),$$

$$K_A(c, m_i) = \begin{cases} d(A(c), A(m_i)) & \text{if both outputs are sense codons,} \\ P_{\text{stop}} & \text{if } A(c) \text{ is sense and } A(m_i) \text{ is stop,} \\ 0 & \text{if } A(c) \text{ is stop.} \end{cases} \quad (2)$$

where $\mu(c, m_i)$ is the transition–transversion weight and $P_{\text{stop}} = 50{,}000$ is a nonsense penalty.

2.3 Total Code Cost

The total cost of an assignment A is the sum of vulnerabilities over all 64 codons:

$$C(A) = \sum_{c \in \text{codons}} V_A(c). \quad (3)$$

2.4 Assignment Space

An **assignment** is a bijection $A\colon \text{AminoAcid} \to \text{AminoAcid}$ that relabels which amino acid each codon family encodes, while preserving the degeneracy pattern (stop codons remain stop codons, and all codons in a synonymous family receive the same label). The identity assignment recovers the standard genetic code.

The **pairwise swap neighborhood** consists of all $\binom{20}{2} = 190$ transpositions (a, b) with $a \neq b$, where the codon families for a and b exchange their amino-acid labels while all other families remain unchanged.

2.5 Local Optimality

The standard code is **locally optimal** under cost function C if $C(\text{id}) \leq C(\sigma_{a,b})$ for every transposition $\sigma_{a,b}$ in the swap neighborhood. It is **not locally optimal** if there exist $a \neq b$ with $C(\sigma_{a,b}) < C(\text{id})$.

3 Results

3.1 The Standard Code Is Not Locally Optimal

Under the chemical-distance metric of Eq. (1), the standard code has total cost $C(\text{id}) = 3{,}581{,}250$.

Of the 190 pairwise swaps, **42 are strictly improving** (reduce total cost). The remaining 148 are strictly worsening; no swap is neutral.

Table 1 lists the 10 largest-improving swaps. The single best swap, Arg ↔ Gln, reduces cost by 367,312 (−10.26%). The top eight improving swaps all involve arginine, indicating that the chemistry-only objective systematically over-penalizes the mutational neighborhood of the largest 6-fold codon family.

Table 1: Top 10 improving pairwise swaps under simple chemical distance. ΔC is the cost reduction relative to the standard code.

Swap	Degeneracy	ΔC	%
Arg ↔ Gln	6×2	−367,312	−10.26
Arg ↔ Asn	6×2	−362,696	−10.13
Arg ↔ Cys	6×2	−329,794	−9.21
Arg ↔ His	6×2	−313,700	−8.76
Arg ↔ Asp	6×2	−253,104	−7.07
Arg ↔ Glu	6×2	−242,232	−6.76
Arg ↔ Lys	6×2	−241,430	−6.74
Arg ↔ Met	6×1	−230,034	−6.42
Glu ↔ Gly	2×4	−155,500	−4.34
Gly ↔ Tyr	4×2	−153,156	−4.28

3.2 The Improving Swaps Are Not Random

The 42 improving swaps show strong structural patterns:

1. **Arg dominates.** 11 of the 42 improving swaps involve arginine, and the top eight improving swaps all involve arginine. The chemistry-only objective assigns arginine to an unusually expensive mutational neighborhood.
2. **Cross-degeneracy swaps dominate.** 30 of the 42 improving swaps exchange amino acids with different degeneracy counts (for example a 6-fold family with a 2-fold or 1-fold family). The metric therefore fails to price the structural consequences of degeneracy asymmetry.
3. **The improving basin is broad.** The top 20 improving swaps include exchanges involving 6-fold, 4-fold, 2-fold, and 1-fold families. This is not a single isolated anomaly but a persistent feature of the chemistry-only landscape.

3.3 Random Baseline

To calibrate the standard code's position in the chemical-distance landscape, we sampled 10,000 degeneracy-preserving random assignments (shuffling amino acid labels within each degeneracy class independently, seed = 42). The standard code sits at approximately the **0.53rd percentile by cost**: only 53 of the 10,000 sampled random codes are as good or better. Equivalently, the standard code outperforms 99.47% of sampled degeneracy-preserving shuffles under this metric.

This is consistent with the Freeland–Hurst finding [2] that the standard code outperforms most random codes, but refines it: the code is globally strong among random alternatives yet still not locally optimal. The point is therefore not that the code is mediocre under simple chemistry, but that random-code comparisons alone do not establish that the right objective has been found.

3.4 Simulated Annealing

Simulated annealing with 500,000 iterations (temperature schedule $T = 10{,}000 \cdot (1/10{,}000)^{t/500{,}000}$, seed = 42) finds codes with cost 2,709,790, which is 24.33% cheaper than the standard code. This is far below the best of the 10,000 sampled random shuffles and confirms that the chemistry-only objective contains a substantial improving basin, not merely a few isolated swap anomalies.

4 Formal Verification

All results are formalized in Lean 4 (version 4.27.0) using the Mathlib library. The key modules and theorems are:

- `chemicalDistance` in `AminoAcidZClassification.lean`: the distance function of Eq. (1), proved symmetric and zero on the diagonal.
- `codonMutationWeight` in `MutationModel.lean`: transition/transversion weighting, proved symmetric and zero on equal nucleotides.
- `totalAssignmentCost` in `CodonAssignmentObjective.lean`: the total cost function of Eq. (3).
- `chemOnly_argGln_improves_nat` in `ComputableAssignmentCost.lean`: a `native_decide` proof that the Arg ↔ Gln swap strictly reduces cost under natural-number arithmetic.
- `chemOnly_not_locally_optimal` in `CertifiedResults.lean`: the theorem that there exist $a \neq b$ with $C(\sigma_{a,b}) < C(\mathrm{id})$.

The full formalization comprises 49 Lean files totaling over 8,000 lines of code, with zero axioms and zero unproved assertions (`sorry`). Every definition is computable; every inequality is verified by the Lean kernel via `native_decide` on natural-number

arithmetic.

5 Discussion

5.1 What the Negative Result Means

The standard genetic code is not locally optimal under a simple chemical-distance metric. This does not mean the code is poorly designed. It means the metric is incomplete. The code was presumably shaped by constraints that a four-property Euclidean distance cannot capture:

1. **Position-dependent mutation effects.** Mutations at the second codon position have far larger chemical consequences than mutations at the third (wobble) position [9]. A metric that weights all positions equally conflates these very different biological costs.
2. **Biosynthetic pathway constraints.** Amino acids within the same biosynthetic family (e.g., the pyruvate-derived set: Ala, Val, Leu) tend to be encoded by neighboring codon boxes. A simple chemical-distance metric assigns no special value to this clustering.
3. **Translation-machinery compatibility.** The aminoacyl-tRNA synthetase system imposes structural constraints on which amino acids can be assigned to which codon families. A metric that treats all reassignments as equally plausible ignores these constraints.
4. **Charge conservation at wobble.** The wobble position (third nucleotide) tolerates synonymous mutations that preserve amino-acid identity but can change codon-level properties such as Z-parity. A metric that does not distinguish wobble from non-wobble mutations cannot capture this asymmetry.

5.2 Relation to Prior Work

The Freeland–Hurst result [2] established that the standard code is better than ∼99% of random codes under a weighted polar-requirement objective. Our result is complementary, not contradictory. Under our chemistry-only objective the standard code also beats 99.47% of sampled degeneracy-preserving random shuffles, yet still fails local optimality. The discrepancy arises because rank among random codes is a global statistic, whereas local optimality asks whether any nearby perturbation improves the objective. A code can be globally strong relative to random alternatives and still not be a local minimum of the wrong cost function.

More recent work by Wnętrzak et al. [7] has shown that position-dependent weighting significantly changes the optimality landscape. Our results are consistent with theirs and provide a formal, machine-checked foundation for the same conclusion.

5.3 Constructive Implications

The negative result is constructive: it identifies exactly which swaps improve the code and by how much. The pattern of improving swaps—dominated by arginine and by cross-degeneracy exchanges—provides direct clues about what a correct optimality metric must penalize.

In a companion paper [Paper 2], we show that extending the metric with position-dependent weighting and biosynthetic pathway constraints produces an objective under which the standard code *is* the unique global optimum over all 363,004 degeneracy-preserving assignments.

6 Methods

6.1 Chemical Properties

Amino acid properties are drawn from standard reference values: molecular weights, Kyte–Doolittle hydrophobicity scale [5] multiplied by 10 and rounded to integers, charge at pH 7 ($\{-1, 0, +1\}$), and side-chain volumes from Zamyatnin [10]. All 20 canonical amino acids are included.

6.2 Computation

The optimizer is implemented in Python 3.12 and available in `tools/codon_assignment_optimizer.py`. All results are deterministic (random seed = 42 where applicable). Frozen outputs are stored in `artifacts/paper1_frozen_results.json`. The random baseline reported in this paper uses 10,000 sampled degeneracy-preserving shuffles; the simulated annealing run uses 500,000 iterations.

6.3 Formal Verification

The Lean 4 formalization is in the `IndisputableMonolith.Genetics` namespace of the `reality` repository. Build command: `lake build IndisputableMonolith.Genetics`. The build requires Lean 4.27.0 and a pre-downloaded Mathlib cache (`lake exe cache get`).

7 Conclusion

The standard genetic code is not locally optimal under a simple chemical-distance metric. Forty-two of 190 pairwise amino-acid swaps strictly reduce the total weighted mutation cost, with the best swap (Arg $\leftrightarrow$ Gln) improving the objective by 10.26%. At the same time, the standard code still outperforms 99.47% of sampled degeneracy-preserving random shuffles. This combination of facts is machine-checked in Lean 4, eliminating the possibility of computational error or off-by-one mistakes in the enumeration.

The finding does not diminish the code's remarkable structure. Instead, it sharpens the question: what objective *does* the code minimize? The answer must include position-dependent mutation effects, biosynthetic pathway constraints, and translation-machinery compatibility—all of which are absent from the simple chemical-distance metric tested here.

Data Availability

All code, data, and Lean source files are available in the `reality` repository. The optimizer script, frozen result tables, and Lean formalization can be used to reproduce every number in this paper.

Acknowledgments

The Lean formalization uses the Mathlib library maintained by the `leanprover-community`. Computational resources were standard desktop hardware.

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